



DRESS SYNDROME

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Drug reaction with eosinophilia and systemic symptoms (DRESS) is a rare hypersensitivity reaction to certain medications that is potentially life-threatening. Multiple studies show the estimated mortality, or death rate, to be about 10%. DRESS syndrome can occur in both children and adults. It is more commonly seen in adults without a preference for sex. The incidence of DRESS varies depending on both an individual's ethnicity and the causative drug. The overall incidence of DRESS syndrome is uncommon, with the estimated risk being 1 in 1,000 to 1 in 10,000 after drug exposure, although this risk also changes based on the offending drug. For example, the estimated incidence for the antiepileptic drug phenytoin was 2.3-4.5 per 10,000.

Patients diagnosed with DRESS syndrome typically present with a rash, fever, and eosinophilia but can have a variety of symptoms including liver, lung, or kidney involvement. Literature review shows that this disease has a latency of about two to six weeks between the drug exposure and the onset of symptoms. The skin findings usually start as a measles like rash that progresses to become more diffuse and red.

It is important to distinguish DRESS syndrome, a serious and severe systemic reaction, from less severe allergic reactions to medications or other causes of rash. Some diseases that may mimic DRESS syndrome include but are not limited to: **Stevens-Johnson syndrome/toxic epidermal necrolysis**, Sézary syndrome, **systemic lupus erythematosus**, **staphylococcal scaled skin syndrome**, and others. If a clinician has a high suspicion of DRESS syndrome, there are findings in the history, physical exam, and lab values that can assist in making the diagnosis of DRESS syndrome and ruling out other diseases.

The diagnosis of DRESS syndrome is complex and often requires evaluation from a physician before ruling out other causes. DRESS syndrome should be suspected if a diffuse rash erupts and is accompanied by fever, facial edema, and enlarged lymph two to six weeks after starting a new high-risk medication. A table of typical or classic findings in DRESS syndrome can be found in Table 1.

<u>History</u>	<u>Skin findings</u>	<u>Systemic symptoms</u>	<u>Organ involvement</u>
Patient began taking high-risk medication 2-6 weeks ago, such as antiepileptic drugs, allopurinol, sulfonamides, dapsone, or other antibiotics	Rose-red or slightly elevated rash that begins to become confluent	Fever, enlarged and tender lymph nodes	Blood: Elevated white blood cells, eosinophils, monocytes
Previous history or DRESS syndrome	Exfoliative dermatitis involving >90% of the body	Lymph nodes appear similar to a viral infection on biopsy	Heart/Lungs: Interstitial pneumonia, pleural effusion, myocarditis
Itching or fever that precedes the rash	Lymphocytes visualized on skin biopsy near blood vessels.	Re-activation of human herpesvirus-6 and other viruses	Other: Renal impairment, elevated liver enzymes, or inflammation/involvement of other organs

Table 1- A list of classic signs or symptoms that a person with DRESS syndrome may experience. Please note that not all are required for diagnosis.

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If a physician suspects DRESS syndrome, it is advisable to investigate further by searching for abnormal lab findings, such as high levels of eosinophils, neutrophils, monocytes, atypical lymphocytes, or elevated liver enzymes. It may be prudent to obtain a skin biopsy to determine if lymphocytes have invaded the affected area. Scoring systems exist to aid in making the diagnosis of DRESS syndrome (Table 2), although there is not a reliable standard that is used or required to make the diagnosis due to the variety of symptoms and findings that someone with DRESS syndrome may experience.

Table 2: Scoring criteria for DRESS syndrome

Fever > 38.5 C		Present		Absent	Unknown
Enlarged lymph nodes (2 sites at least)		0		-1	-1
Eosinophilia: 700-1499 or 10-19.9 percent		1		0	0
Eosinophils >1500 or >20%		2		0	0
Skin rash >50%		1		0	
At least 2: Edema, infiltration, purpura, scaling		1		-1	
Biopsy suggesting DRESS		1		-1	
Internal organ involvement:	Two or more organs	1	2	0	0
One					
Resolves in >15 days		0		-1	-1
Evaluation for other causes (autoantibodies, hepatitis, chlamydia, mycoplasma serology, etc.) with negative results of >3		1		0	0

Table 2- A scoring system originally described by Kardaun et. al to predict the likelihood of a diagnosis of DRESS syndrome based on total number of points. < 2 = excluded, 2-3 = possible, 4-5 = probable > 6 = definite Kardaun SH, Sidoroff A, Valeyrie-Allanore L, et al. Variability in the clinical pattern of cutaneous side-effects of drugs with systemic symptoms: does a DRESS syndrome really exist? Br J Dermatol. 2007;156(3):609-611

While the exact etiology of DRESS syndrome is not entirely understood, it is known that a strong immune response to specific drugs is a key factor. This strong immune reaction involves an early increase in T cells of the immune system and has been shown to reactivate viral infections that were previously latent in the affected individual, including many viruses of the herpesvirus family, including infectious mononucleosis. 40% of DRESS cases have been shown to be associated with infectious mononucleosis. Studies have shown that several human leukocyte antigen (HLA) haplotypes, a genetic factor, are associated with increased risk of DRESS syndrome.

Once DRESS syndrome has been diagnosed, patients with DRESS syndrome are advised to discontinue immediately the drug inducing the reaction. **Systemic steroids** are often required if there is severe lung, liver, or kidney involvement. If the organ involvement is mild, patients may be treated symptomatically, often with high-dose topical corticosteroids. Patients with severe skin involvement are often hospitalized and require fluid and electrolyte replenishment. Because anti-epileptics are often a source of DRESS syndrome, patients may be switched to valproic acid to provide prophylaxis against seizures. It is important to be aware of the signs, symptoms, and associated risk factors of DRESS syndrome, as early diagnosis and treatment greatly improve outcomes.

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